

THE TWELFTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

March 22, 1952

Morning Session

1. Let

$$f(x) = \sum_{i=0}^{i=n} a_i x^{n-i}$$

be a polynomial of degree n with integral coefficients. If a_0 , a_n , and $f(1)$ are odd, prove that $f(x) = 0$ has no rational roots. (page 340)

2. Show that the equation

$$(9 - x^2) \left(\frac{dy}{dx} \right)^2 = (9 - y^2)$$

characterizes a family of conics touching the four sides of a fixed square. (page 340)

3. Develop necessary and sufficient conditions which ensure that r_1, r_2, r_3 and r_1^2, r_2^2, r_3^2 are simultaneously roots of the equation $x^3 + ax^2 + bx + c = 0$. (page 345)

4. The flag of the United Nations consists of a polar map of the world, with the North Pole as center, extending approximately to 45° South Latitude. The parallels of latitude are concentric circles with radii proportional to their co-latitudes. Australia is near the periphery of the map and is intersected by the parallel of latitude 30° S. In the very close vicinity of this parallel how much are East and West distances exaggerated as compared to North and South distances? (page 349)

5. Let $a_j (j = 1, 2, \dots, n)$ be entirely arbitrary numbers except that no one is equal to unity. Prove

$$a_1 + \sum_{i=2}^n a_i \prod_{j=1}^{i-1} (1 - a_j) = 1 - \prod_{j=1}^n (1 - a_j). \quad (\text{page 350})$$

6. A man has a rectangular block of wood m by n by r inches (m, n , and r

are integers). He paints the entire surface of the block, cuts the block into inch cubes, and notices that exactly half the cubes are completely unpainted. Prove that the number of essentially different blocks with this property is finite. (Do *not* attempt to enumerate them.) (page 351)

7. Directed lines are drawn from the center of a circle, making angles of $0, \pm 1, \pm 2, \pm 3, \dots$ (measured in radians from a prime direction). If these lines meet the circle in points $P_0, P_1, P_{-1}, P_2, P_{-2}, \dots$, show that there is no interval on the circumference of the circle which does not contain some $P_{\pm i}$. (You may assume that π is irrational.) (page 353)

Afternoon Session

1. A mathematical moron is given two sides and the included angle of a triangle and attempts to use the Law of Cosines: $a^2 = b^2 + c^2 - 2bc \cos A$, to find the third side a . He uses logarithms as follows. He finds $\log b$ and doubles it; adds to that the double of $\log c$; subtracts the sum of the logarithms of 2, b , c , and $\cos A$; divides the result by 2; and takes the anti-logarithm. Although his method may be open to suspicion, his computation is accurate. What are the necessary and sufficient conditions on the triangle that this method should yield the correct result? (page 353)

2. Find the surface generated by the solutions of

$$\frac{dx}{yz} = \frac{dy}{zx} = \frac{dz}{xy},$$

which intersects the circle $y^2 + z^2 = 1, x = 0$. (page 354)

3. Develop necessary and sufficient conditions that the equation

$$\begin{vmatrix} 0 & a_1 - x & a_2 - x \\ -a_1 - x & 0 & a_3 - x \\ -a_2 - x & -a_3 - x & 0 \end{vmatrix} = 0 \quad (a_i \neq 0)$$

shall have a multiple root. (page 356)

4. A homogeneous solid body is made by joining a base of a circular cylinder of height h and radius r , and the base of a hemisphere of radius r . This body is placed with the hemispherical end on a horizontal table, with the axis of the cylinder in a vertical position, and then slightly oscillated. It is intuitively evident that if r is large as compared to h , the equilibrium will be stable; but if r is small as compared to h , the equilibrium will be unstable. What is the critical value of the ratio r/h which enables the body to rest in neutral equilibrium in any position? (page 357)

5. If the terms of a sequence, a_n , are monotonic, and if $\sum_1^\infty a_n$ converges, show that $\sum_1^\infty n(a_n - a_{n+1})$ converges. (page 359)

6. Prove the necessary and sufficient condition that a triangle inscribed in an ellipse shall have maximum area is that its centroid coincide with the center of the ellipse. (page 360)

7. Given any real number N_0 , if $N_{j+1} = \cos N_j$, prove that $\lim_{j \rightarrow \infty} N_j$ exists and is independent of N_0 . (page 361)